MIL-P-20385E
2 December 1982
SUPERSEDING
MIL-P-20385D
22 February 1977

MILITARY SPECIFICATION

PROTRACTORS, SEMICIRCULAR, PLASTIC, GRADUATED IN MILS, YARDS, METERS, AND INCHES

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers semicircular and plastic protractors.

1.2 Classification. Protractors shall be the following types and styles, as specified, (see 6.2):

Type I - Graduated in mils, and yards or meters, 1:25,000, and 1:50,000 scale.
- Style 1 - 12-inch diameter, graduated in yards.
- Style 2 - 12-inch diameter, graduated in meters.
- Style 3 - 16-inch diameter, graduated in meters.

Type II - Graduated in mils, yards and inches, or meters and centimeters.
- Style 1 - 10-inch diameter, graduated in yards and inches.
- Style 2 - 10-inch diameter, graduated in meters and centimeters.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-535	- Plastic Sheet (Sheeting): Plastic Strip: Poly (Vinyl Chloride) and Poly (Vinyl Chloride-Vinyl Acetate), Rigid.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-1882	- Preparation for Delivery Requirements for Templates, Triangles, Curves, Protractors and Related Drafting Aids.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The protractors shall be as shown in figures 1 through 5, and as specified herein.

3.2 Materials. Materials shall be as specified herein (and on the drawings). Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

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3.2.1 Plastic. Plastic shall conform to L-P-535, composition B, type I, class 2.

3.3 Graduations, numerals, and markings. Graduations, numerals, and markings shall be molded, impressed, or printed on the underside of the protractor and filled or printed with permanent black color, except as specified in 3.6, so as to be readable from the upper side. Lettering and numerals shall be a readily legible size and style.

3.3.1 Graduations. When tested as specified in 4.4.2.1, graduation lines shall be within 0.010 inch of their true relative position. All graduation lines shall be 0.008 inch wide, plus or minus 0.002 inch.

3.4 Laminations. If the graduations and markings on the lower surface of the protractor are printed, they shall be protected by a transparent plastic cover sheet between 0.003 and 0.005 inch thick of the same material as the protractor. The cover sheet shall be permanently affixed by a laminating process which shall not impair the transparency of the protractor or cover sheet, nor distort the markings or graduations. When tested as specified in 4.4.2.2, there shall be no evidence of the cover sheet separating from the protractor.

3.5 Identification marking. Each protractor shall be identified in accordance with MIL-STD-130. All protractors shall be marked with the letters "US" located as shown on the figures.

3.6 Type I, graduated in mils, and yards or meters, 1:25,000 and 1:50,000 scales. Protractors shall be fabricated as shown in figures 1, 2, and 3 for styles 1, 2, and 3, respectively, and as specified herein. The 1:50,000 scale and any markings related thereto shall be made with a permanent red color.

3.7 Type II, graduated in mils, yards and inches, or meters and centimeters, 1:10,000, 1:20,000, and 1:40,000 scales. The protractors shall be fabricated as shown in figures 4 and 5 for styles 1 and 2, respectively, and as specified herein.

3.8 Workmanship. The protractors shall be free from warpage, cracks, scratches, bubbles, imbedded particles, and other defects that affect appearance or which might affect serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).

b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level II.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 1.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The protractors shall be examined as specified herein for the following defects:

Major

101. Nonconformance to figures.
102. Material not as specified.
103. Lettering, numerals, or graduations missing or not easily legible.
104. Numerals or graduations not as specified.
105. Cover sheet not as specified.
106. Workmanship not as specified.

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Minor

201. Identification marking missing, incomplete, or illegible.
202. Color or lettering not as specified.

4.4.2 Tests.

4.4.2.1 Accuracy. The protractor shall be measured at not less than four different intervals on the yard or meter range scale, and not less than three different intervals on the protractor mil scale. Nonconformance to 3.3.1 shall constitute failure of this test.

4.4.2.2 Lamination test. The protractor shall be flexed around a cylindrical mandrel with a radius of 8 inches, once face-up and once face-down on the mandrel. The protractors shall then be examined for evidence of delamination. Any evidence of delamination shall constitute failure of this test.

4.5 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with the quality assurance provisions of PPP-P-1882 and the exceptions specified in section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Level A. Protractors of like description shall be preserved in accordance with the level A packaging requirements of PPP-P-1882, except the box for intermediate packaging shall be class weather resistant.

5.1.2 Level B. Protractors of like description shall be preserved in accordance with the level B packaging requirements of PPP-P-1882, except the intermediate packaging shall be as specified in 5.1.1 for level A.

5.1.3 Commercial. Protractors shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Protractors shall be packed together in accordance with the level A packing requirements of PPP-P-1882 except the box shall conform to PPP-B-601, overseas type, style optional. The gross weight of each box shall not exceed 200 pounds. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Protractors shall be packed together in accordance with the level B packing requirements of PPP-P-1882, except the boxes shall be V3c or V3s.

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5.2.3 Commercial. Protractors, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The protractors are intended for use in machine gun and artillery fire control.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and style of protractor required (see 1.2).

c. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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Air Force - 99

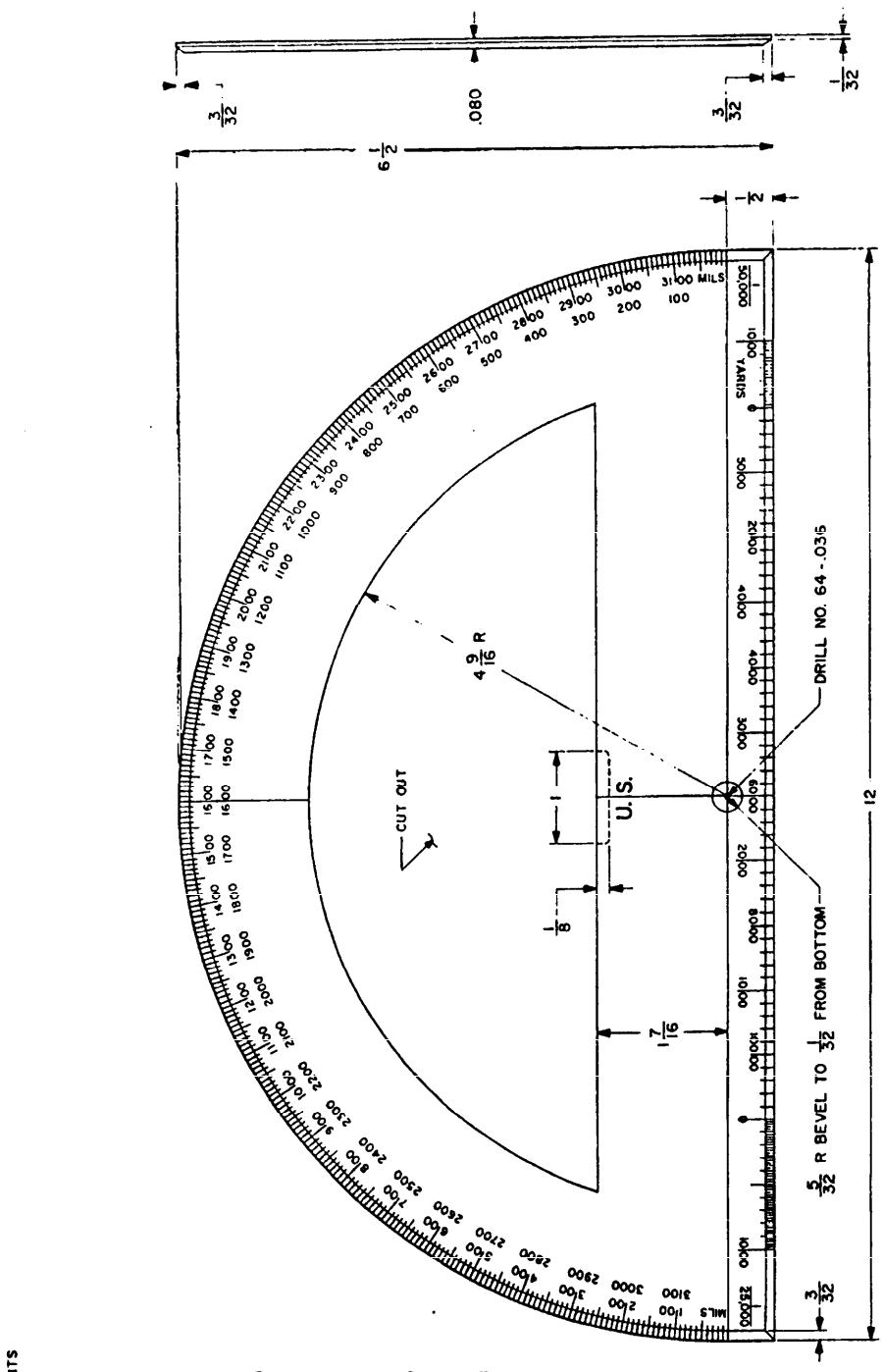
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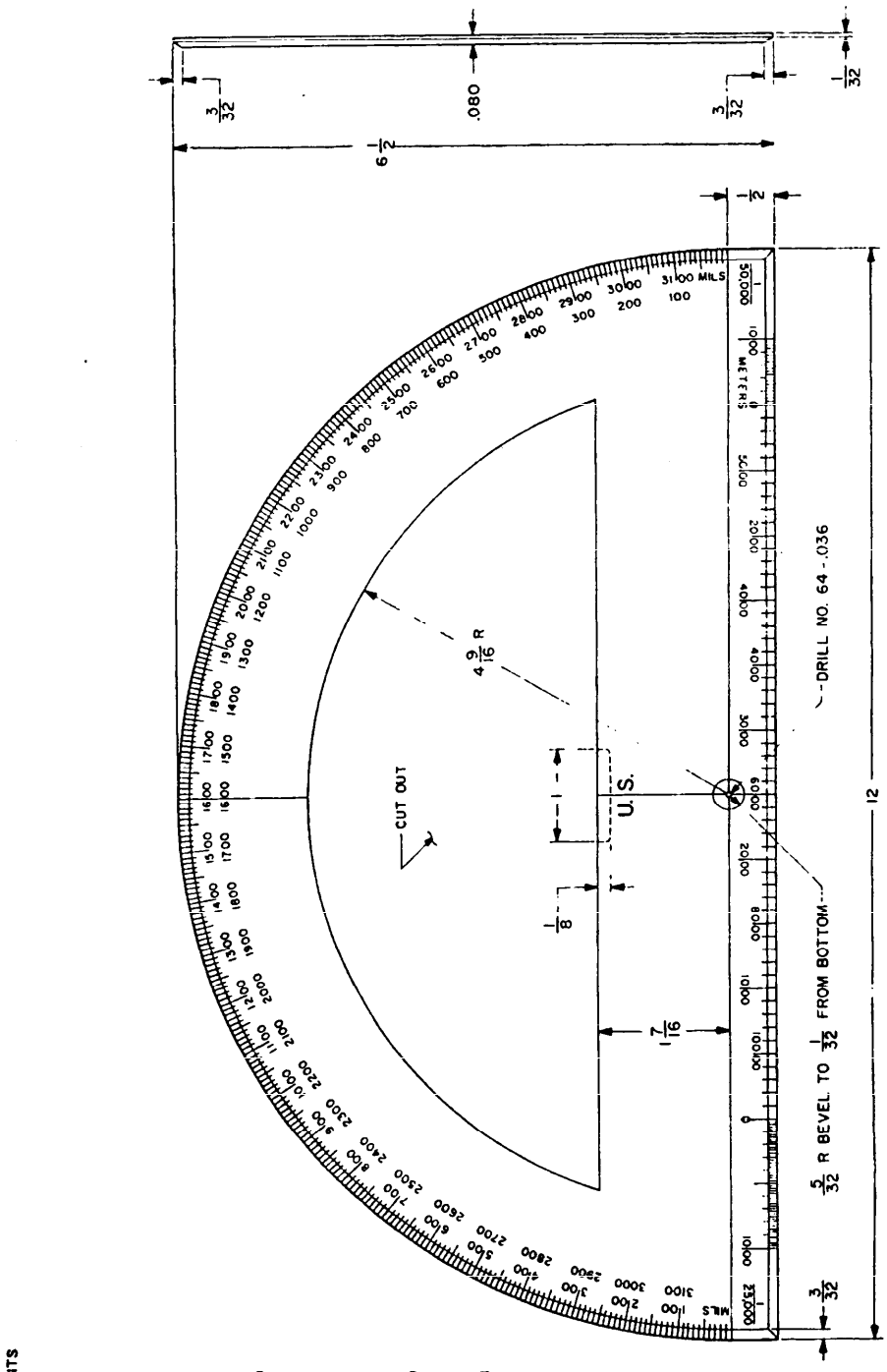
METRIC EQUIVALENTS

INCHES	MM
1/32	.8
.036	.9
.080	2.0
3/32	2.4
1/8	3.2
5/32	4.0
1/2	12.7
1	25.4
1 7/16	36.5
4 9/16	115.9
6 1/2	165.1
12	304.8

FIGURE 1. Protractor, scales 1/25,000 and 1/50,000, 12-inch diameter, yards.

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METRIC EQUIVALENTS

INCHES	MM
1/32	.8
.036	.9
.080	2.0
3/32	2.4
1/8	3.2
5/32	4.0
1/2	12.7
1	25.4
1 7/16	36.5
4 9/16	115.9
6 1/2	165.1
12	304.8

FIGURE 2. Protractor, scales 1/25,000 and 1/50,000, 12-inch, diameter, meters.

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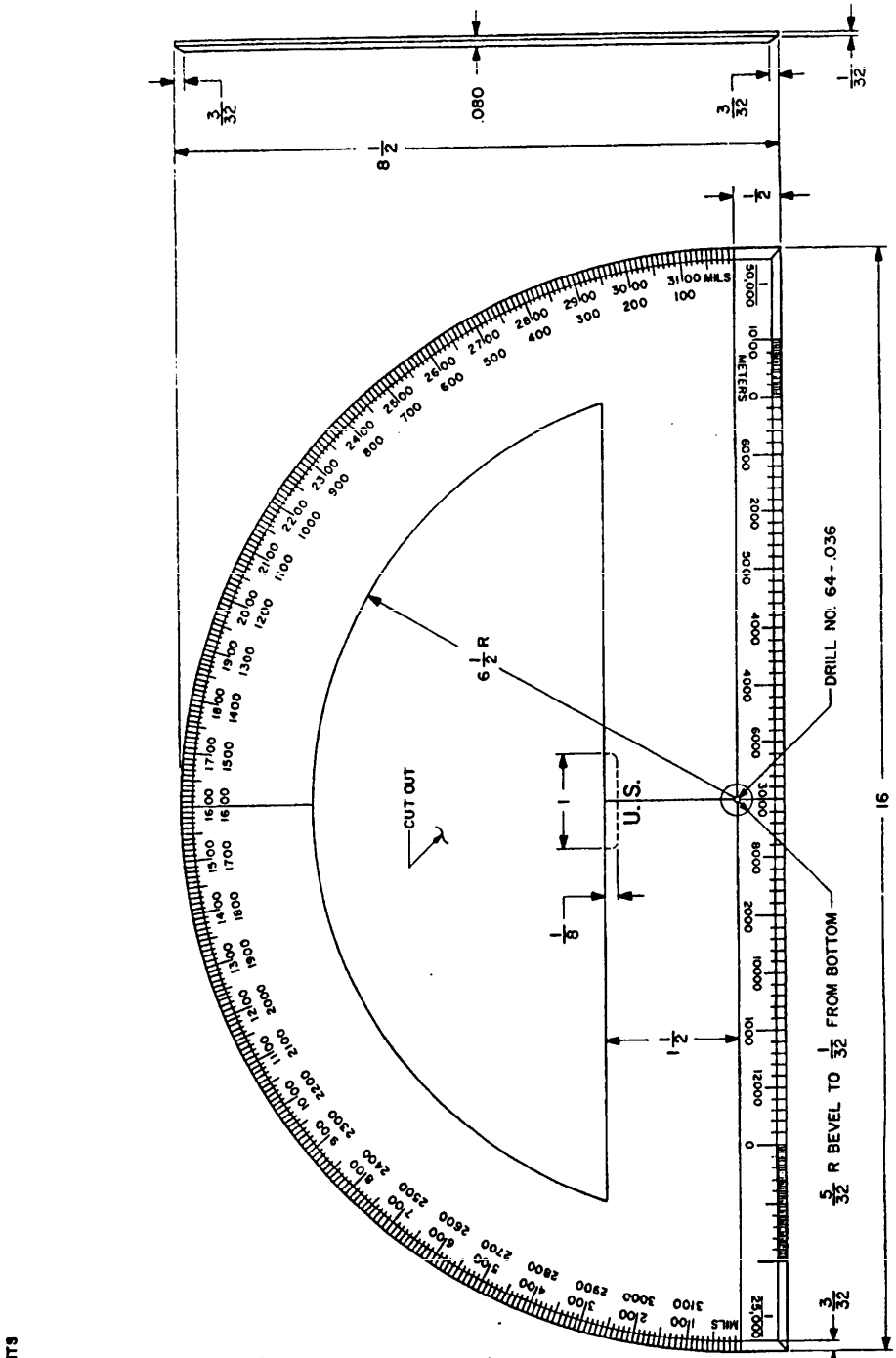
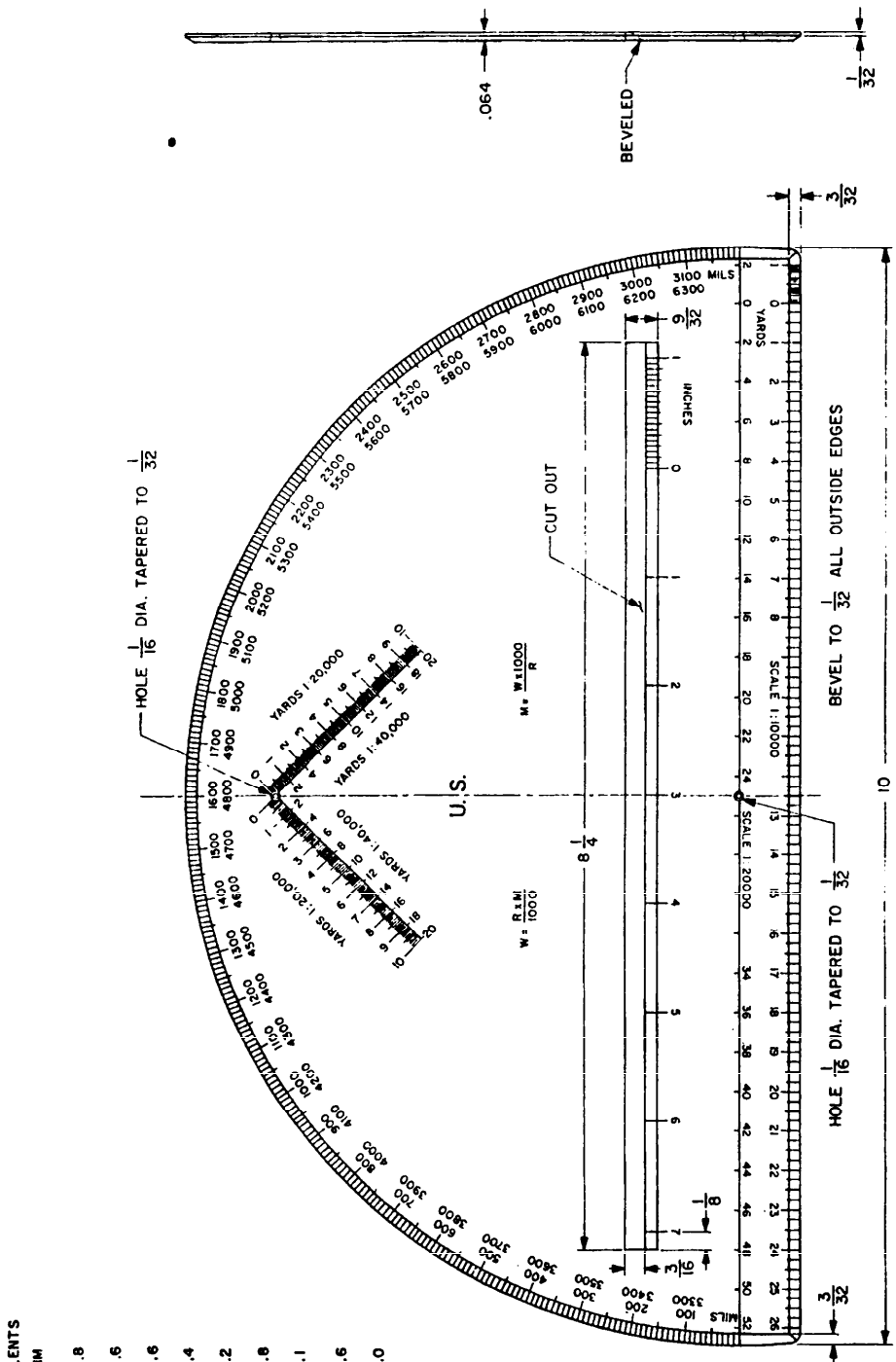


FIGURE 3. Protractor, scales 1/25,000 and 1/50,000, 16-inch diameter, meters.

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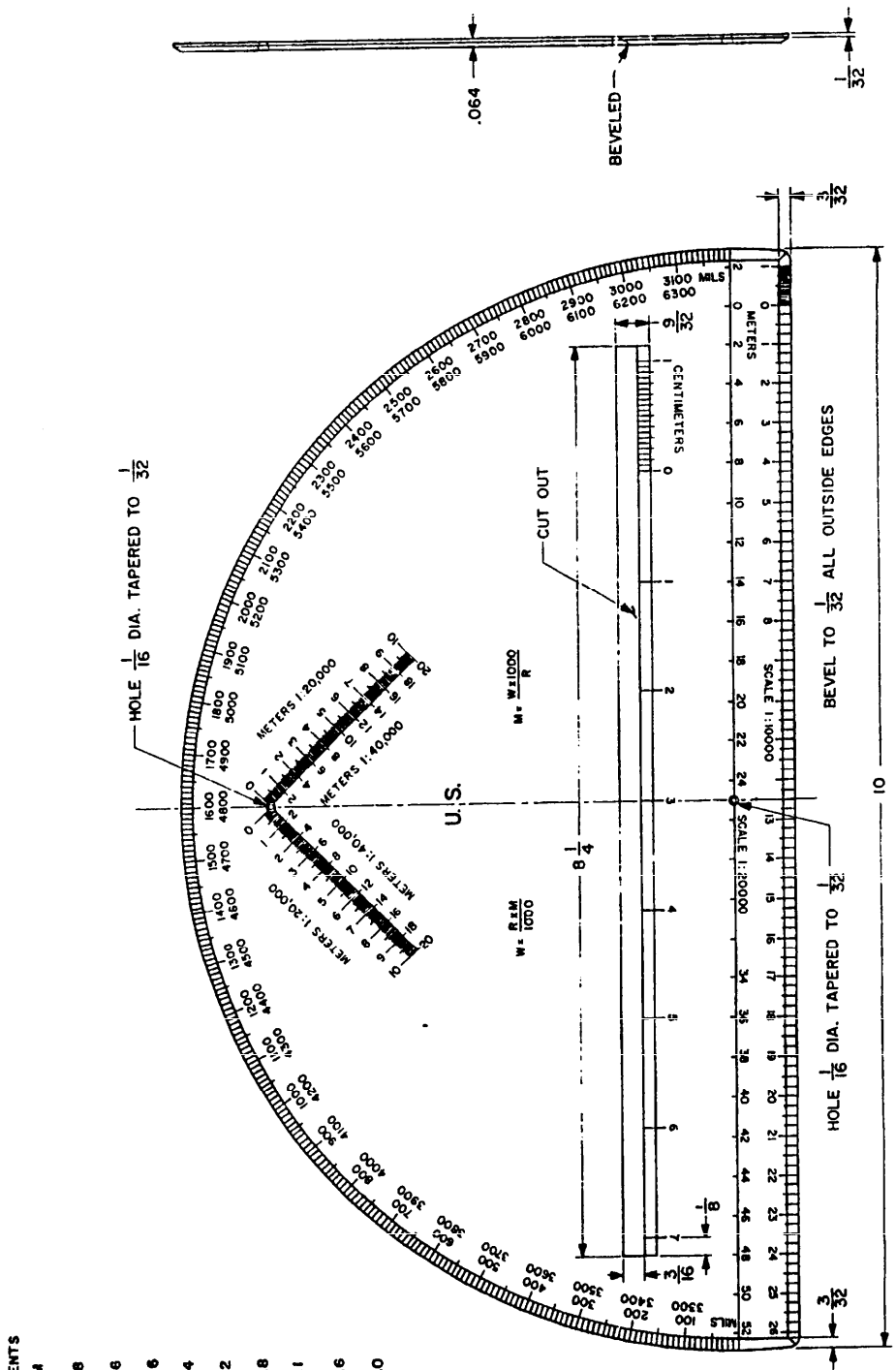
METRIC EQUIVALENTS

INCHES	MM
1/32	.8
1/16	1.6
.064	1.6
3/32	2.4
1/8	3.2
3/16	4.8
9/32	7.1
8 1/4	209.6
10	254.0

FIGURE 4. Protractor, scales 1/10,000, 1/20,000 and 1/40,000, 10-inch diameter, yards and inches.

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METRIC EQUIVALENTS

INCHES	MM
1/32	.8
1/16	1.6
.064	1.6
3/32	2.4
1/8	3.2
3/16	4.8
9/32	7.1
8 1/4	209.6
10	254.0

FIGURE 5. Protractor, scales 1/10,000, 1/20,000 and 1/40,000, 10-inch diameter, meters and centimeters.

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